



# Lamiss Ben Haj Saad

Software Engineer

☎ +216 52 127 818 | 🌐 My Portfolio | ✉ haj.saad.lamiss@gmail.com  
in /in/lamis-ben-haj-saad | 🐙 github.com/Lamis-haj-saad

Curious and adaptable Software Engineering graduate with experience in AI, data science, and multiple programming languages. Passionate about building data-driven solutions and continuously exploring emerging technologies.

## Education

### Software Engineering – Software Architecture

Sep 2022 – Jul 2025

Higher Institute of Applied Sciences and Technology of Sousse

- **Relevant Coursework:** Software Development Lifecycle (SDLC), Software Architecture and Design, Artificial Intelligence, Machine Learning, Computer Vision, Data Structures & Algorithms, Database Systems, Data Science, Cloud Computing, DevOps, Mobile Development, Web Technologies, Programming Languages and Frameworks.

### Pre-engineering: Maths, Physics, Informatics

Sep 2020 – Jun 2022

Higher Institute of Applied Sciences and Technology of Sousse

- **Relevant Coursework:** Data Structures and Algorithms, Prob & Stats, Linear Algebra, Analysis, Networking, System Administration, System Design, Object Oriented Programming.

## Internships & Experience

### Full Stack Developer – Internship

Feb 2025 (4 Months)

#### DECADE

*Project: Design, development and integration of an AI-based assistant into a catering services platform.*

- Completed an accelerated training in ReactJS, Magento, Elasticsearch and E-commerce.
- Analyzed functional and technical requirements to define specifications for an intelligent assistant.
- Developed the client interface using **ReactJS** and built a **Node.js** server to process API requests.
- Implemented speech transcription using Speech-to-Text models to process voice input.
- Integrated NLP modules to interpret user intentions and generate personalized recommendations.
- Applied **prompt engineering** to optimize model responses and improve recommendation accuracy.
- Researched and documented the internal architecture of conversational agents (Transformer models, NLU/NLG).

#### Methodologies:

- **Scrum** for iterative development and sprint planning.
- **Layered Architecture:** The Node.js server executes intelligent models (NLP, Transcription), while the interface uses Elasticsearch for direct data access.
- **Prompt Engineering** to optimize model responses and improve the accuracy of recommendations.

#### Stack:

**Front-end:** ReactJS, TypeScript, SCSS

**Back-end:** Node.js, JavaScript, Express.js, REST API

**Data & Communication:** JSON, Axios

**Search Engine:** Elasticsearch

**AI Models:** vosk-model-fr-0.22, OpenHermes-2.5-Mistral-7B-GGUF

### Hangfire Service Developer – Internship

Aug 2024 (1 Month)

#### TWINKLE-IT

*Project: Design and development of an automated task scheduling service.*

- Designed and developed a background job scheduling service, enabling users to create and schedule tasks.
- Integrated **Hangfire** to manage background tasks within the service, eliminating the need for manual task triggering.
- Transformed the service into an API enabling background task management.

**Stack:** C#, ASP.NET Core, Hangfire, Entity Framework, PostgreSQL

### Front-End Developer – Internship

Aug 2022 (1 Month)

#### E-DIGIAPP

*Project: Development of a showcase website for a metal curtain production company.*

- Translated UI designs into responsive, dynamic web pages.

**Stack:** HTML, CSS, jQuery, JavaScript

## Projects

### ISSATSO+ – Web Platform for ISSAT Sousse

2 Months

Team: 3 People

- Built discussion forums enabling communication between students and teachers.
- Digitized administrative procedures including registration and document requests.
- Developed a quiz and test module allowing teachers to create assessments for student evaluation.
- Automated attendance management through a **signature recognition system** using digital fingerprints.

**Methodologies:** Waterfall methodology, MVC (Model-View-Controller) architecture.

**Stack:** HTML, CSS, JavaScript, jQuery, Bootstrap, PHP 8.3

### Traffic Prediction – Data Science & Machine Learning

2 Weeks

- Cleaned and prepared traffic data : engineered time and speed features to improve model performance.
- Conducted exploratory data analysis to visualize traffic patterns by hour and day.
- Built and evaluated a **Random Forest Regressor** using MAE and RMSE metrics.

**Stack:** Python, Pandas, NumPy, Scikit-learn, Matplotlib

### Digit Recognition System – Computer Vision

2 Weeks

- Built an image processing pipeline (grayscale, Gaussian filtering, Canny edge detection) for digit extraction.
- Trained a **KNN** classifier on 5,000 digit samples; overlaid predictions directly on source images.

**Stack:** Python, OpenCV, NumPy

### Distributed Banking Management System

1 Month

Team: 2 People | Methodology: Waterfall

- Implemented data distribution across multiple sites.
- Enforced secure access control to protect sensitive banking data.
- Managed real-time data updates and automated transactions.

**Stack:** Advanced SQL, functions, cursors, relational-object databases, stored procedures, triggers

### WhatsApp Clone – Mobile Application

1 Month

- Implemented user profile creation, viewing, and updating.
- Built private and group instant messaging with real-time status updates (online/offline).
- Enabled sending of text messages and image files.

**Stack:** JavaScript, React Native, CSS

### Connect Four Game

2 Weeks

- Implemented game logic, player turn management, score tracking, and real-time updates.

**Stack:** Python, Pygame, Tkinter

## Skills

|                |                                                                                                                                                                                  |
|----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Languages      | English (fluent), French (fluent), Arabic (native)                                                                                                                               |
| Programming    | JavaScript, TypeScript, ReactJS, React Native, Node.js, Express.js, PHP, Symfony, Spring Boot, C#, .NET, XAF Blazor, Java (OOP, JavaFX), Android (SQLite), C/C++, Python, MATLAB |
| Databases      | MySQL, PostgreSQL, PL/SQL, Oracle, Elasticsearch, Data Modeling, Data Warehousing, Power BI                                                                                      |
| AI & ML        | Machine Learning, Computer Vision, Prompt Engineering, Scikit-learn, Pandas, NumPy                                                                                               |
| Cloud & DevOps | AWS (Fundamentals), Git, GitHub/GitLab, Linux, DevOps Fundamentals                                                                                                               |
| Methodologies  | Agile/Scrum, UML, Software Architecture, REST APIs, Web 2.0 & Web 3.0                                                                                                            |
| Soft Skills    | Problem Solving, Social Intelligence, Effective Communication, Team Collaboration, Adaptability, Strong Memory, Attention to Details                                             |

## Volunteering

Cyber Trace ISSATSo (Financial Team Member)

Sep 2021 – May 2022

Google Developpers Students Club (Member)

Sep 2021 – Jul 2025

ARSII (Member)

Sep 2023 – Jul 2024

## Certificates

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|------------------------------------------------------------------|----------|
| • Exploring SAP Datasphere – SAP                                 | Jul 2026 |
| • Building AI Agents with Google ADK – Datacamp                  | Jun 2026 |
| • Introduction to Machine Learning – Kaggle                      | Sep 2024 |
| • Python – Kaggle                                                | Aug 2024 |
| • Pandas – Kaggle                                                | Aug 2024 |
| • Agile Project Management – Google                              | Jun 2024 |
| • Building Database Applications in PHP – University of Michigan | Apr 2024 |